SECTION A: FUNDAMENTALS

1. Database is data that is structured and organised, this collection of data can be easily accessed, managed and updated, such as online banking or airline reservation system.

2. Data warehouse is a large and centralised repository used to store all data, historical and current data. The data used is collected from multiple operational systems.

3. Data lake is a storage system that is used to hold large amounts of raw data, in its original format.

4. Data mart is a focused subset of a datawarehouse that is designed for a specific business department or function such as sales.

5.

Structured	Highly organised, data that fits neatly ~~in a~~ into rows and colums.
Semi-structured	Organisational properties but the type of data does not ~~com~~ conform to a rigid table
Unstructured	No predefined format, usually text-heavy or contains binary data

6. The process of creating a visual representation of

~~an~~ If of an information system to show how data is structured.

7 The process of automatically discovering hidden patterns correlations and trends.

8 Thus a software that acts as an interface between users / applications and database allowing them to define create, query and update the data.

9. Extract, Transform, load is a three step data intergration process used to move data from multiple sources.

10. CSV, JSON and Avro.

CSV (Comma-Separated Values)	A simple, row based plain text format. where each line represents a row and values
JSON (Java Script Object Notation)	is a lightweight textbased semi structured format used in pairs and nested structures

SECTION B: Artifical intelligence + Machine learning

11. Artifical intelligence (AI) is the board feild of computer science dedicated to creating machines that can perform tasks that require human intelligence.

12. Machine learning is a subjet of AI that gives computers the ability to learn from data without being explicitly programmed for every for every rule.

13. Deep learning is a specilised subfeild of machine learning that uses artifical neural networks with many layers to learn complex patterns from large amounts of data.

15. A large language model is a type of deep learning model trained on an enormous amount of text data to understand and generate

~~14 A large~~

14. Natural language Processing is a the branch of AI that gives computers the ability to understand interpet and generate or respond to ~~humane~~ human language

16. Refers to a category of Artifical intelligence that can create new content.

17. The collection of labelled ~~example~~ examples that we feed into a machine learning model. This helps it learn patterns and relationships

18. A feild of AI that enables machines to

interpret and understand visual information from the world. Examples include photographs videos and live camera feeds.

SECTION C: API & Data Exchange.

19. An API is a set of rules and protocols that allow one software application to talk to another. It defines the kinds of requests that can be made.

20. An API key is a unique string of letters and numbers that act as an indentifier and a ~~secret~~ secret token, this string is ~~usally~~ long and makes request to an API.

21. A Rest API is an API that follows the architectural principles of Rest. Rest means Representional State transfer

22. JSON is a lightweight, text-based language, independent data format. this language is easy for ~~re~~ hus humans to read and write. JSON is commonly used in APIs.

SECTION D: TOOLS & BEST PRACTIES.

23 Prompt engineering is the practice

SECTION D :

23 The practice of designing and refining the input text given to large language model in order to get the most accurate output

24 is a higher-order function that takes two arguments And Applies a function to every element

25 The on-demand delivery of computing services - servers storage and databases networking.

26 VCS is a software that traces changes to files overtime allowing us to revert to older, compare changes and collaborate.

27 A Jupyter Notebook is an open source, web-based interactive computing environment that allows us to create documents containing live code, visualizations explanatory text and multimedia elements.

28. Python is a high level programming language known for its simple, readable syntax. It has become the most popular language for Data Analytics.

29 It is an aspect of information technology that deals with the ability of an organisation to collect store and share personal information.

30. Data ethics is the branch of ethics that studies and evaluates moral problems related to data. Examples of ~~this~~ this include: facial recognition misuse.

31 Data governance is the overall management of the availability, usability, intergity and security of data used in an organisation

32 It is an AI system that produces systematically predjuiced results due to assumptions in the machine learning process.

33. The difference is the roles comparison.
Data Analyst deal with Dashboards, SQL Tableau while Data Scientist: predictions ML and Python. Data Engineer deal and work with pipelines, spark, cloud while AI Engineer: Deploy models, Dockers and Kubernetes.

34. It is a technology-Driven process for collecting intergrating and analysing business information. Examples of popular tools include: Power BI and Tableau.

SECTION E

35 I found Generative AI and the idea of large Language modell the most interesting.

36 Deep learning and Neural Networks!
Youtube was helpful in understanding how this works.

37 I believe that Data and AI will deeply affect my career and my everyday life. I already see the effects of AI powered recommedations on netflix, youtube instagram etc.